

You must ask questions to get help

... but it's too scary 🥹

Teacher : Don't be afraid to ask questions

Me: *asks a question*

Teacher :



[Laughing] Young fool

I'm Alex

- My goal: **Teach you how to get help fast while looking good** 😎
- High-performing tech lead at Course Hero, Meta, Robinhood
- \$85k TC -> \$750k TC in 7 years
- Mentored dozens of engineers to ⚡ fast promotions (while answering their *many* questions)



Junior 🧐

Meta

E3

\$192,702

Alex's Mentorship 🌿

3 years

5+ Engineers

Senior 😎

Meta

E5

\$475,444

Greatly Exceeds
Expectations 🚀

*I have raised multiple new-grads to tech leads!

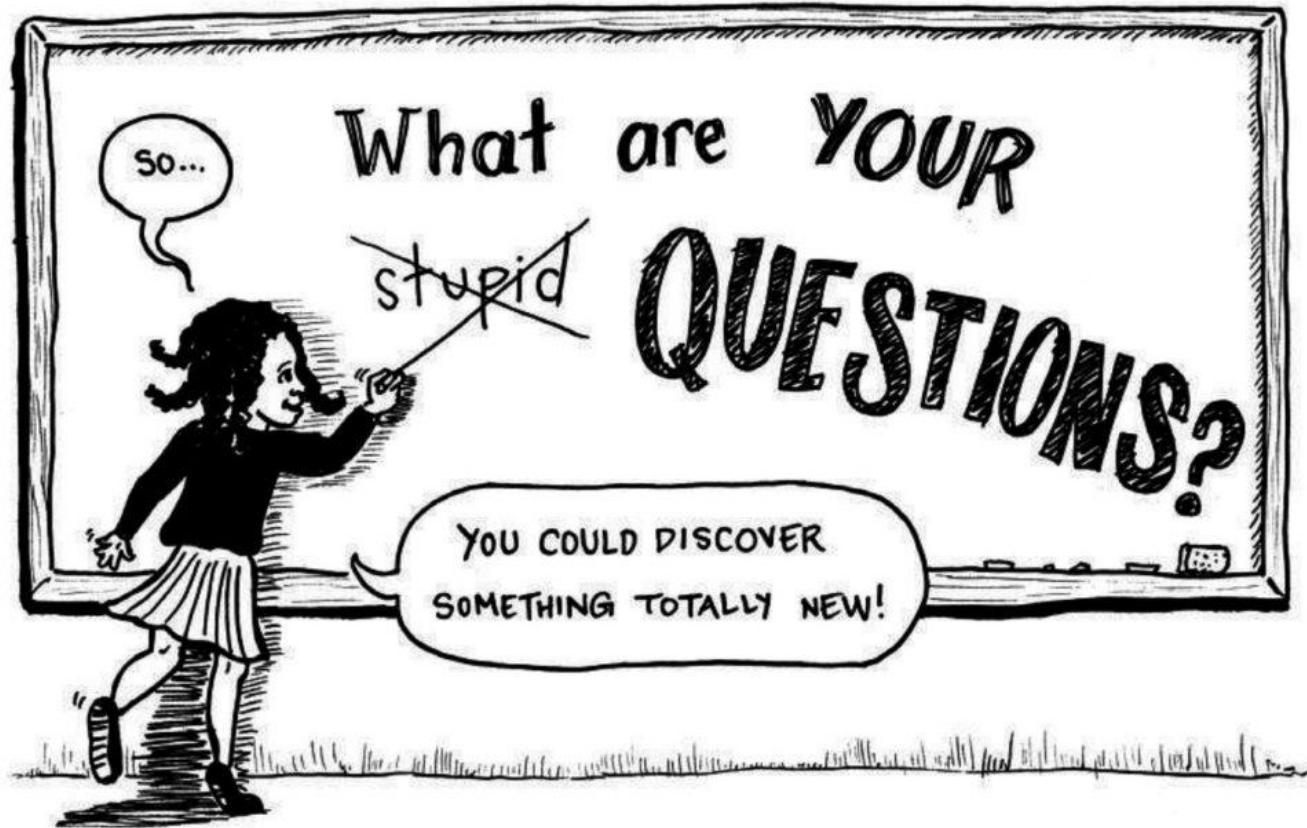


Asking questions is *not* a binary choice 🧑
It's a skill issue and a core marker of seniority 🧠

Objectives

-  Have the right **mentality** behind asking questions
-  Turn questions into a **strength**, not a weakness
-  Understand the difference between a **high-quality** question and a **low-quality** question
-  Ask stellar **technical** questions
-  Ask for **favors** effectively

There Are No Stupid Questions





There are naive questions, tedious questions, ill-phrased questions, questions put after inadequate self-criticism. But every question is a cry to understand the world. There is no such thing as a dumb question.

— *Carl Sagan* —

AZ QUOTES



Software engineering
is *very* information-
intensive 🙄

You need to acquire
info *fast* in order to
survive 🏃

Ask a *lot* of questions.

When Questions Go Wrong 🙄

Good core
(positive
intent)

Low question quality



When Questions Go Right 👍

Good core
(positive
intent)

High question quality



The Key Behind Questions That Don't Suck



EMPATHY



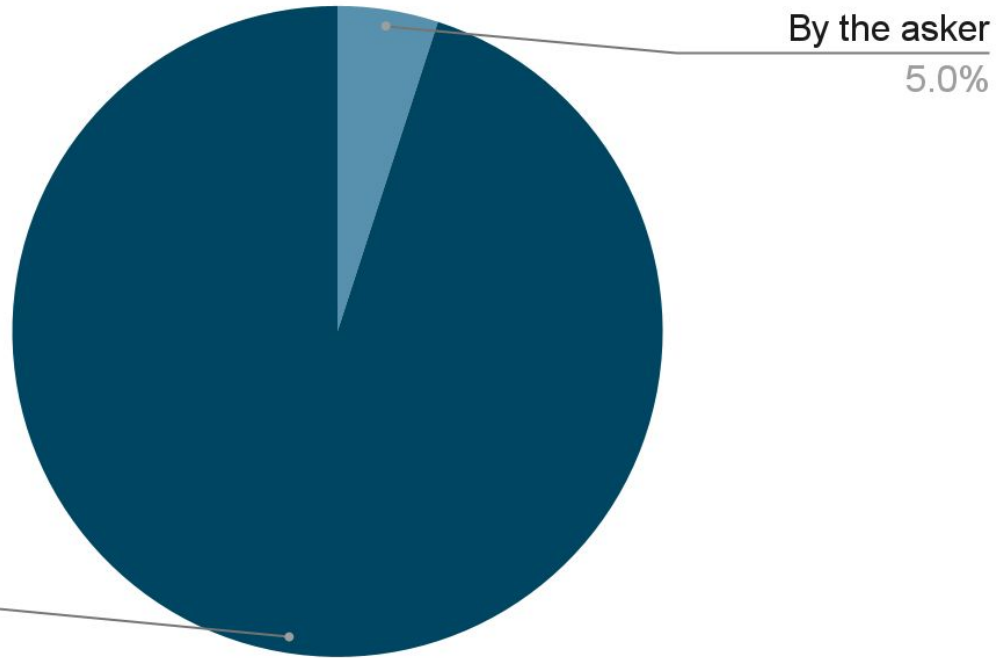
Why Questions Are High-Stakes

- *Questions are fundamentally an exchange of time*
 - The answerer's time is generally more valuable than yours
- Bad question: Robs helpers of their time
- Good question: Respects others' time



Anatomy Of A Bad Question 🙄

Time Spent

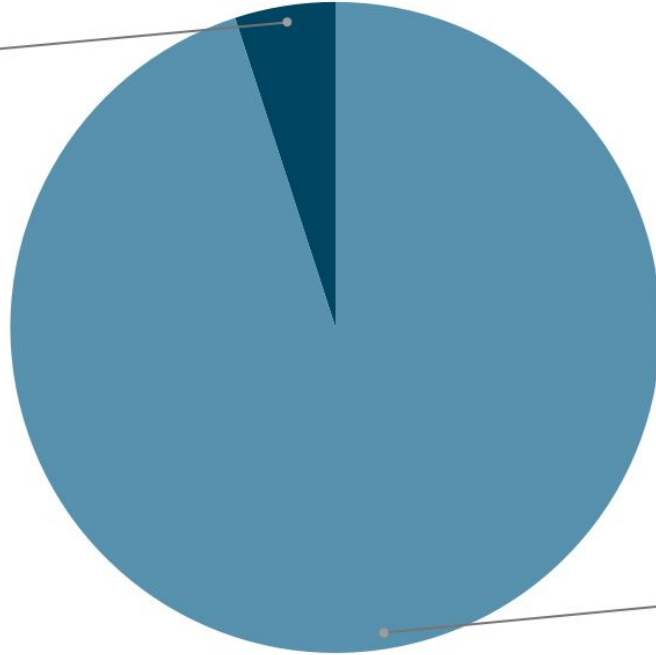


Anatomy Of A Good Question 🧐

Time Spent

By the answerer

5.0%



By the asker

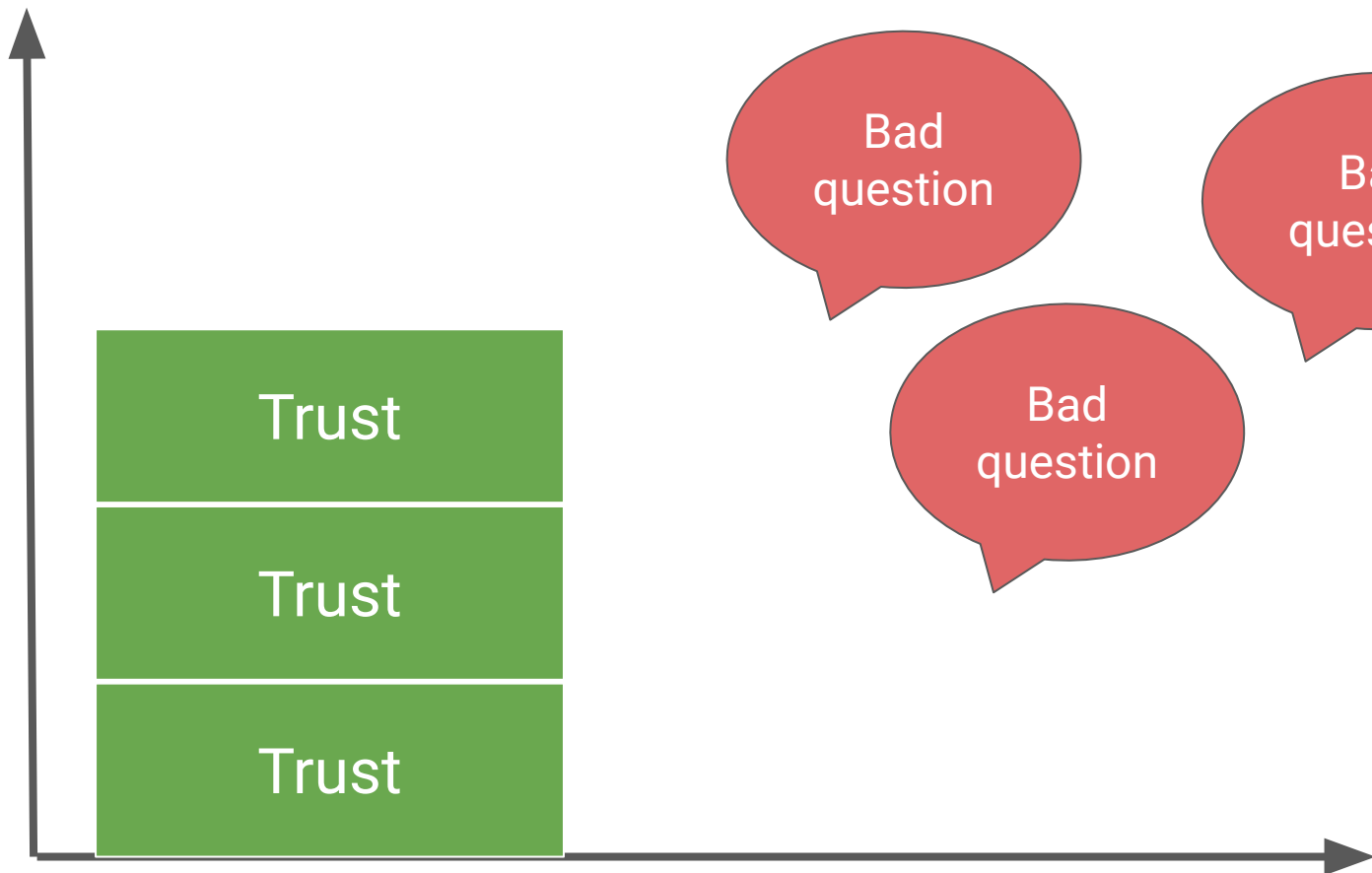
95.0%

**Put yourself in the shoes of the
answerer.**

Make Questions Your Superpower



Your
reputation



Trust

Trust

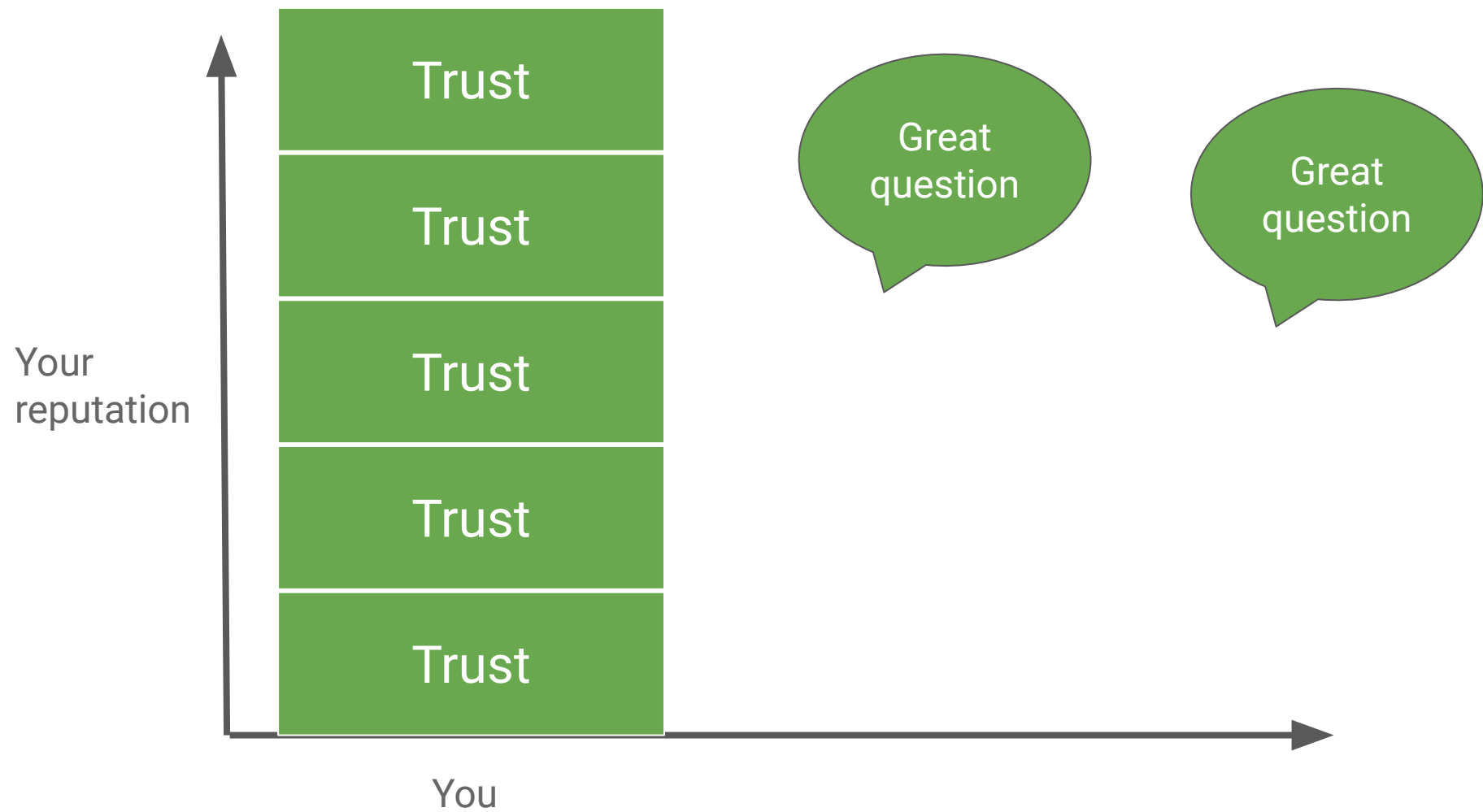
Trust

Bad
question

Bad
question

Bad
question

You



Level 3: Ask amazing questions (great answers, make people want to actively mentor you)



Level 2: Ask good questions (good answers, reputation steady or gradually increasing)



Level 1: Ask bad questions (0 or bad answers, reputation loss)



Level 0: Don't ask questions



This is how Alex found his first mentee at Meta!



How To Build Trust With Questions 💪

- 1 Show in-depth knowledge
- 2 Create a helpful artifact
- 3 Demonstrate humility
- 4 Share positive energy and gratitude

Figure It Out Or Get Help?



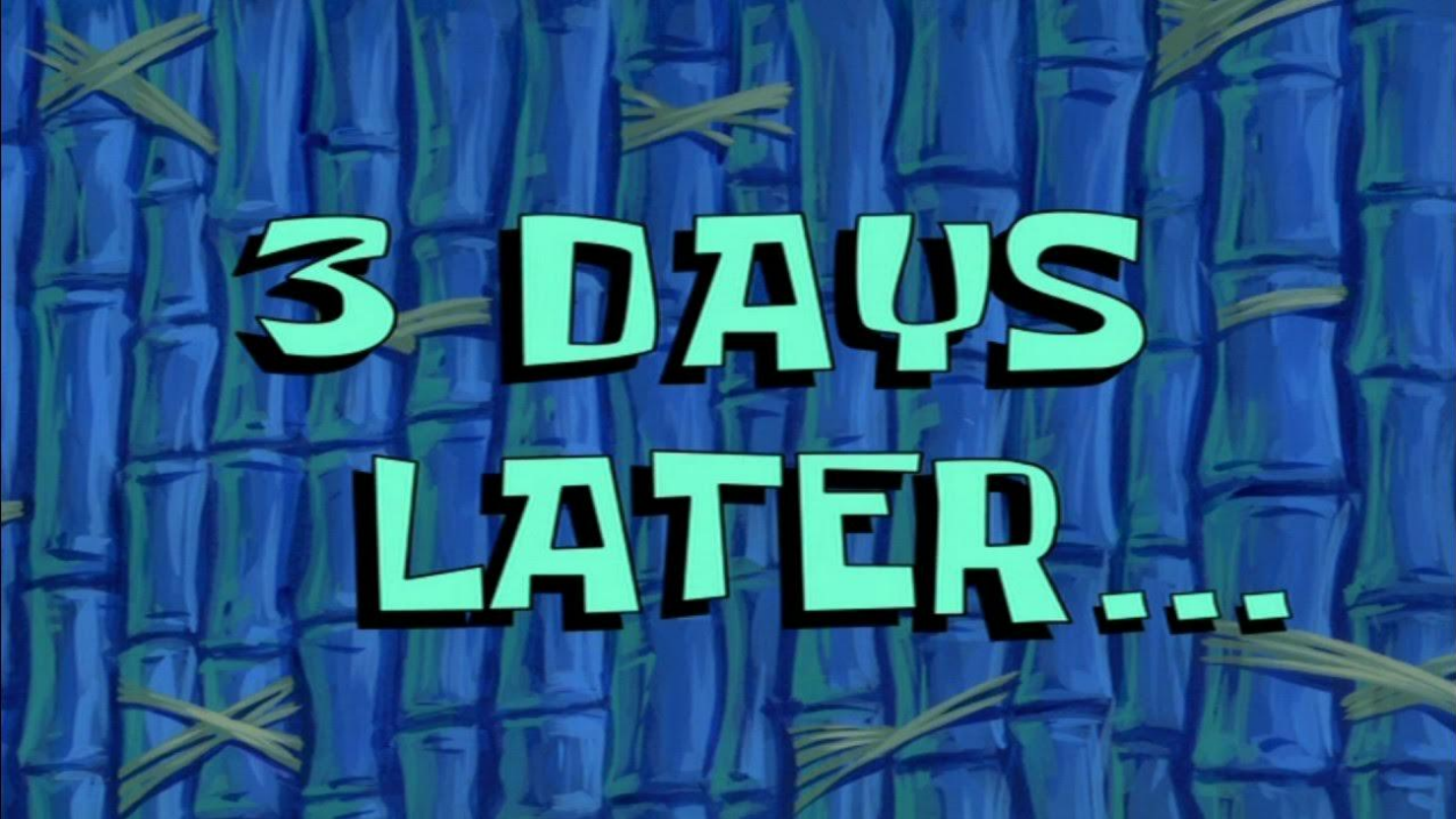


n00b
Engineer

"This task is hard, but I
don't want to bother
anybody and look
stupid. I'll figure it out
on my own..."

Big scary
task

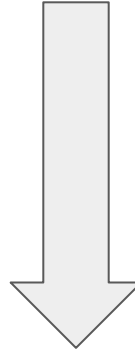




**3 DAYS
LATER...**

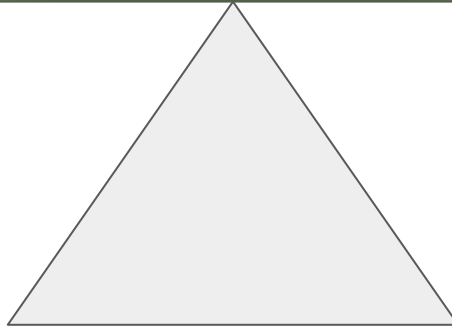


Never ask for
help, struggle
constantly,
burn out, fall
behind 😵

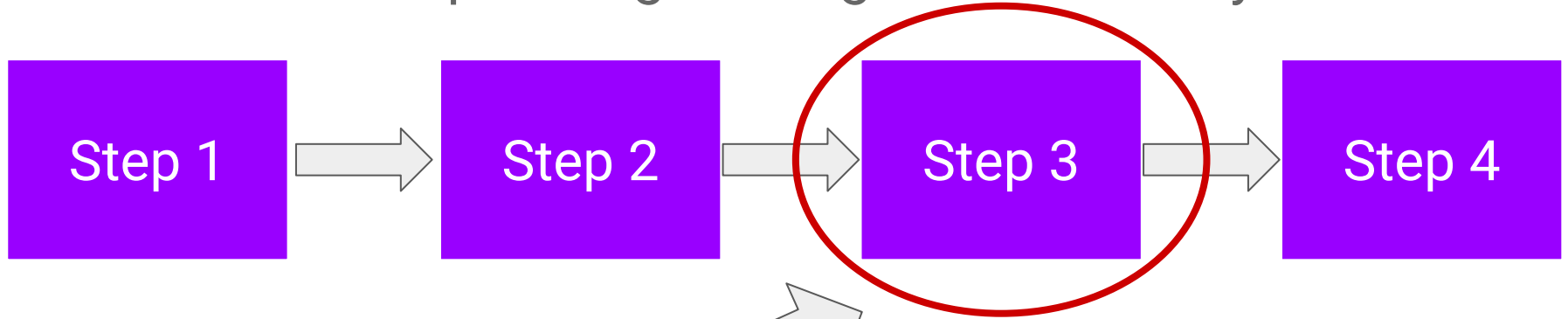


We want to be here ⚖️

Ask for help at
the slightest
sign of struggle,
learn nothing,
burden your
team 😞



How Complex Engineering Problems Play Out



You ask for help when you've
been *stuck* for a while 🧱

A while = 15 - 60 minutes ⌚

Get Help Earlier Or Later?

Earlier

- Urgent deadline
- You are new
- You are junior
- Distributed team
- Your team is open



Later

- Lots of time remaining
- You are tenured
- You are senior
- Commonly located team
- Your team is closed



Be Clueless In Public



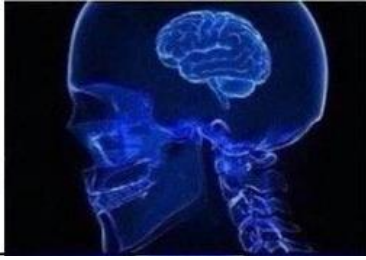
Be Open

- Many engineers hide their lack of knowledge as much as possible
 - Often by relegating question to 1 on 1s
- Do the opposite:
 - Maximize the audience
 - Create an artifact

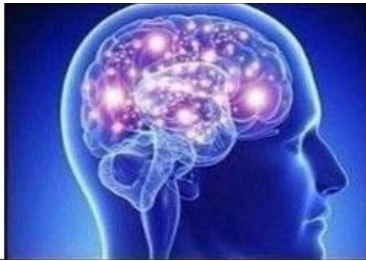


I don't know.

**PRIVATE 1
ON 1 MEETING**



**1 ON 1
CHAT
(SLACK/TEAMS)**



**TEAM
CHAT/GROUP**

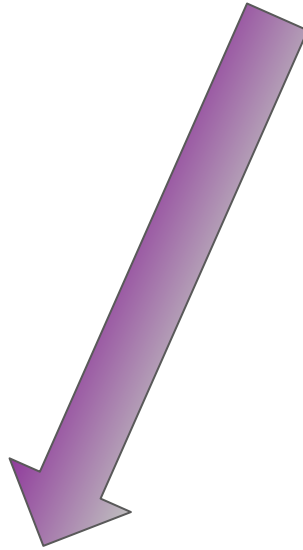


**ORGANIZATION
CHAT/GROUP**



Be here as much as you
possibly can 🗣️

Trickier problems often
need to be run by more
people 💡





You

*"Can someone help
me with this problem
I've been stuck on for
way too long?"*

*"OMG, I'M
STUCK ON THE
EXACT SAME
THING!!!"*



You'll be surprised at how many
people are clueless like you 

Having the courage to ask is
enough to become a leader 

Please Create An Artifact 🖋️



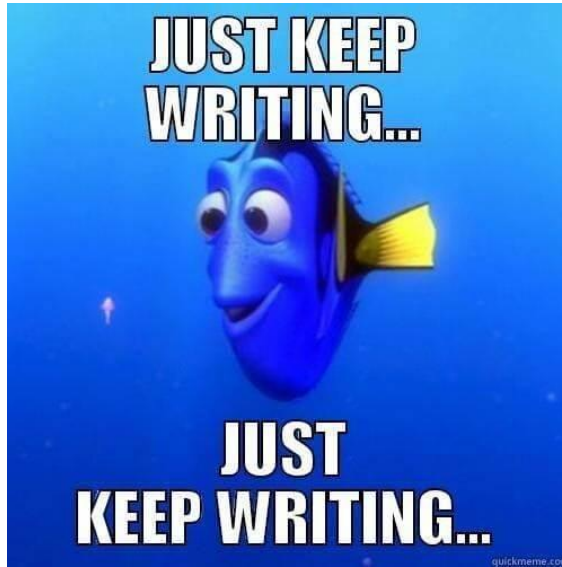
You can reference it later on



It will help current and future teammates



You can point to it as relevant context (e.g. your PR)

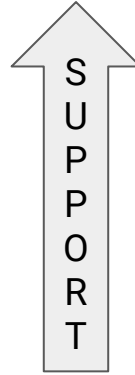


Diversify Your Asks

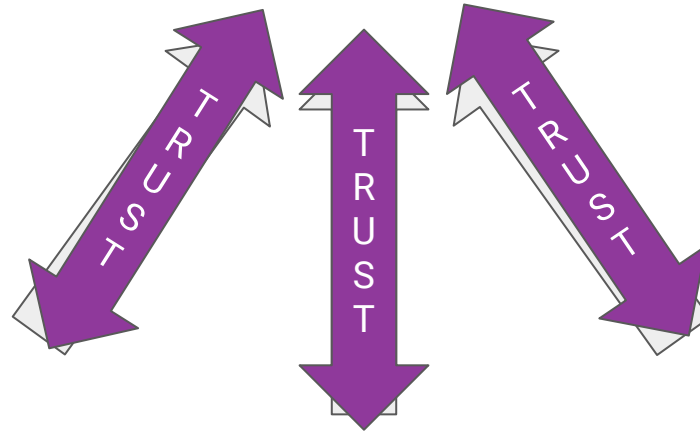


As an engineer, you need
to be careful and avoid
creating *single points of
failure* 🛑

This applies with both
code and people 💡



Teammate

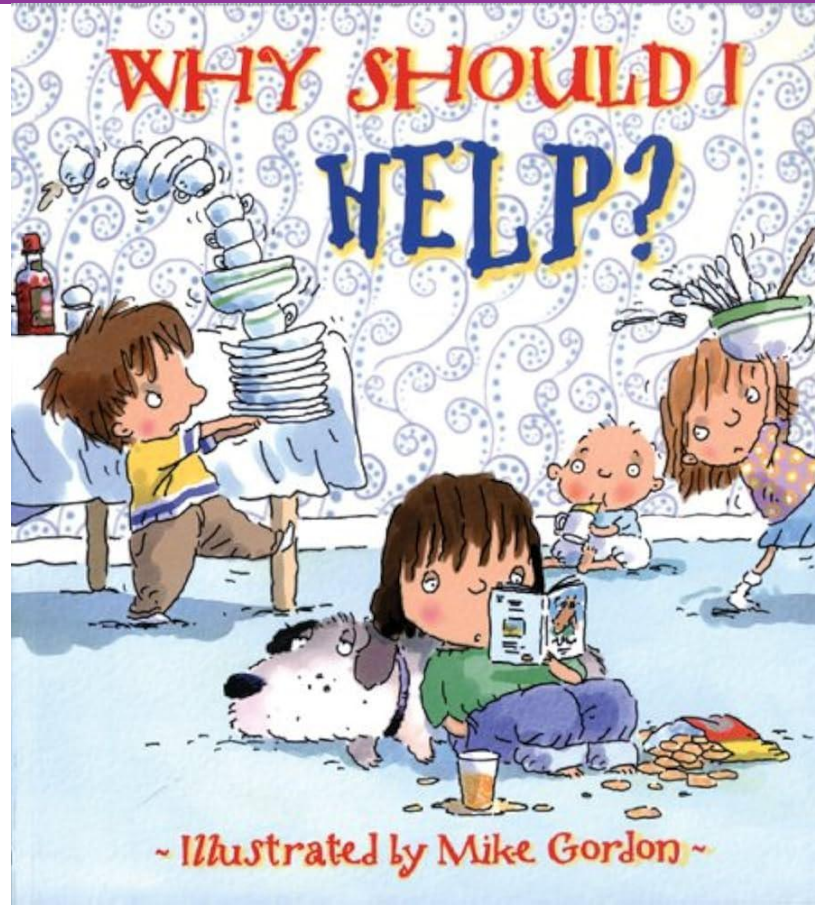


Teammate

Teammate

Teammate

Explain The Impact

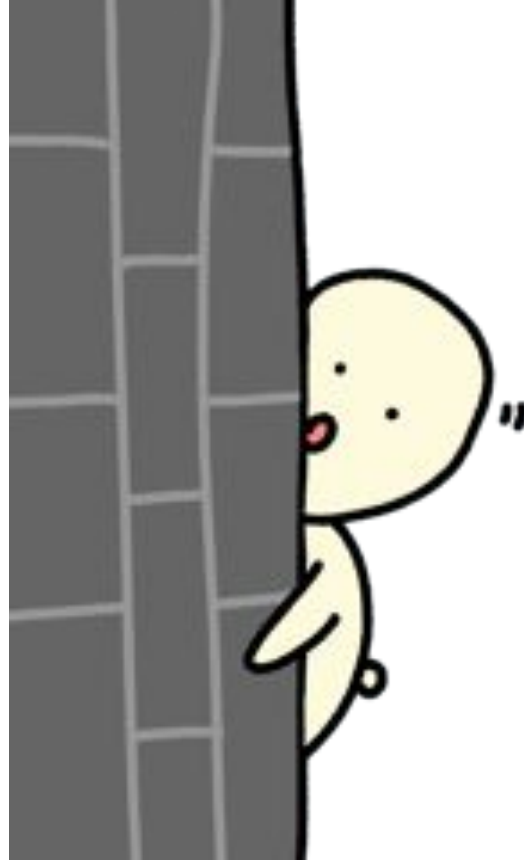


Pictured: An Actual Meta Engineer



"What's the impact?"

"What's the impact?"



?????

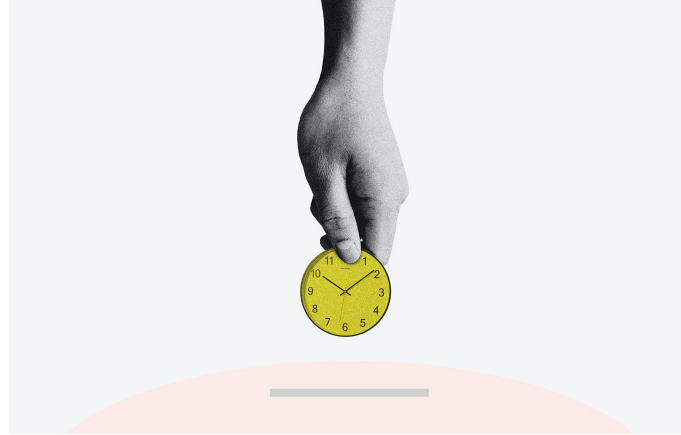
"What's the impact?"

"What's the impact?"

"What's the impact?"

People Want Their Actions To Be Meaningful

- The helpers can only help so many people
 - They need to ruthlessly prioritize
- A major deciding factor behind who to help is the *impact* of their problem
 - So show it!
- *What great thing is unlocked after you're helped?*



*"I'm getting this crash: <UGLY
STACK TRACE SPEW HERE>*

Please help."



*"I'm working on project X, **which is one of my team's Top 3 priorities (+\$50M revenue)**. It's 95% done, but I'm currently stuck on this nasty crash...*

<Bug context tastefully added here>"



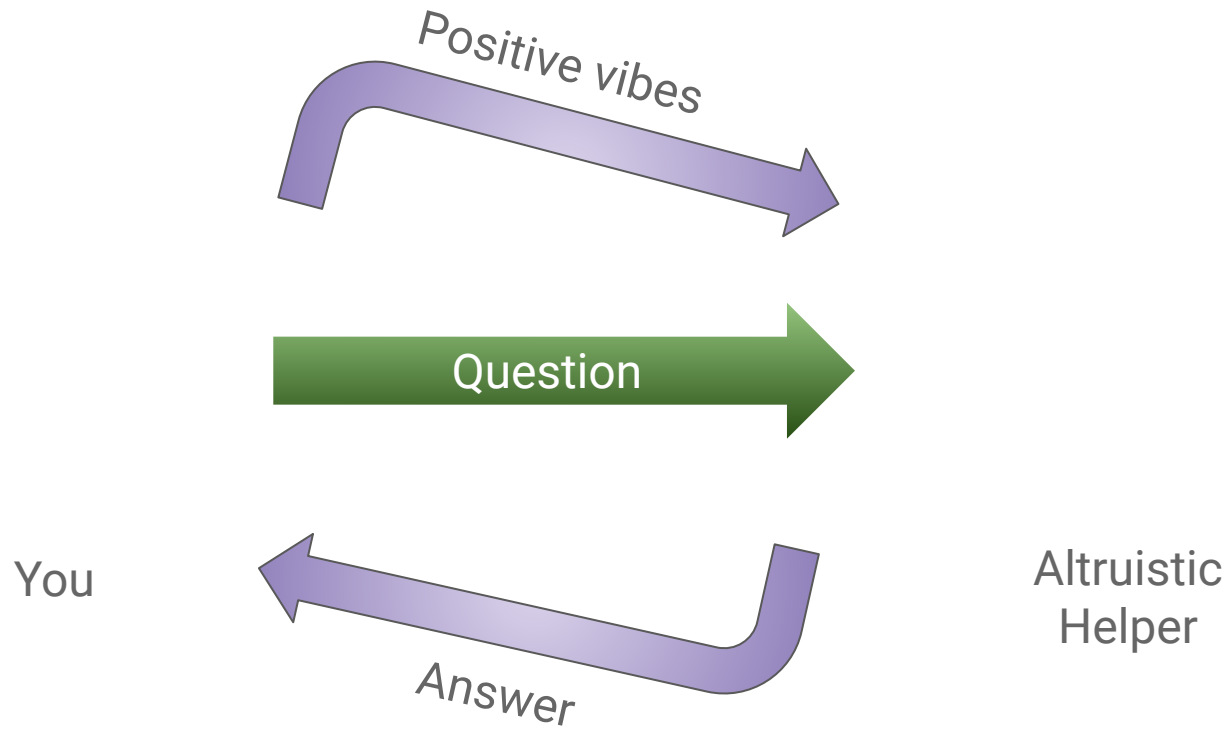
Bonus: Illustrate *Emotional* Impact

*"I'm working on project X, which is one of my team's Top 3 priorities (+\$50M revenue). It's 95% done, but **I've been stuck on this nasty crash for 3 days straight** 😭"*

<Bug context tastefully added here>"

Be Appreciative





"Here's this extremely complicated issue I'm running into: <MOUNTAIN OF HARD TO READ CODE HERE>

Please help."



"<Tastefully added context about the problem (impact, isolated code, relevant links)>

Big thanks for reading through all this - I know it's very complicated and I appreciate you 🙏



"Here's this extremely complicated issue I'm running into: <MOUNTAIN OF HARD TO READ CODE HERE>

Please help. TIA"



"<Tastefully added context about the problem (impact, isolated code, relevant links)>

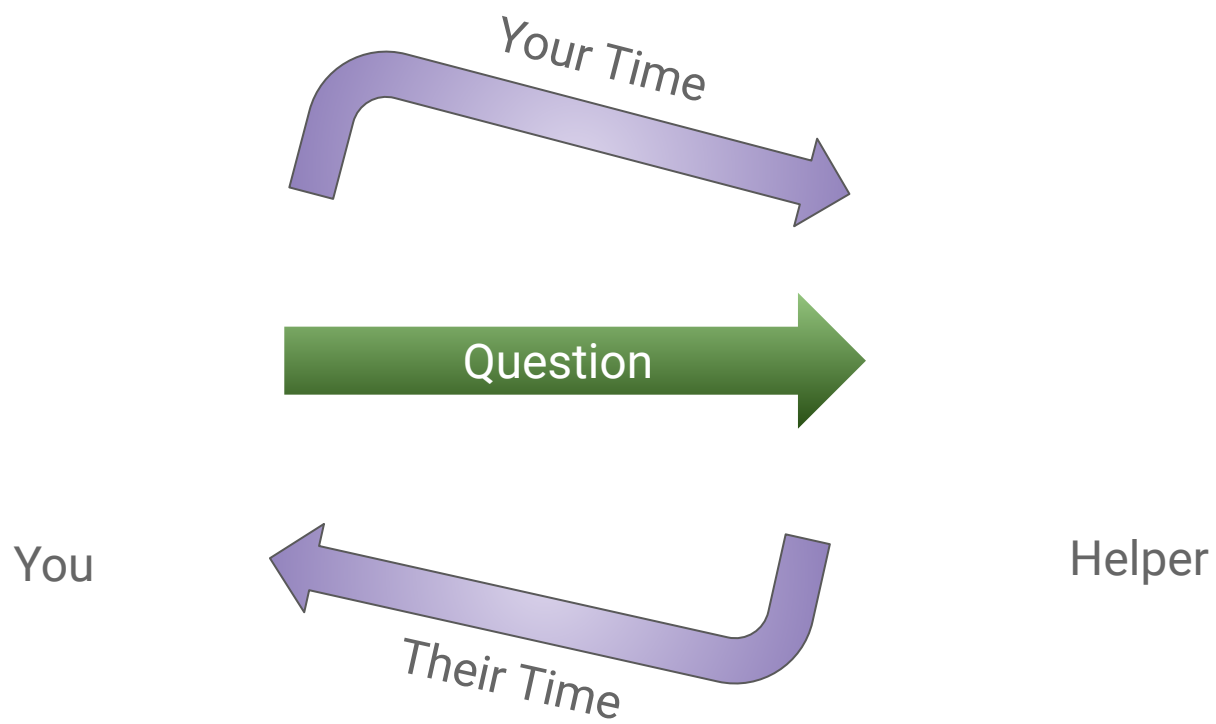
Big thanks for reading through all this - I know it's very complicated and I appreciate you 🙏"

Don't be lazy - Write it out



Put Yourself On The Line





"Here's this extremely complicated issue I'm running into: <MOUNTAIN OF HARD TO READ CODE HERE>

Please help."



"<Tastefully added context about the problem (impact, isolated code, relevant links)>

If I'm missing any relevant context, just comment on this post and I'll add it ASAP!"



Extreme Option: Offer Synchronous Time

When to use?

- The issue is *very* complicated
- You have a good relationship with targeted helper(s)
- Urgency

"<Tastefully added context about the problem (impact, isolated code, relevant links)>

If you need more context, I'm happy to hop on a call and walk you through it!"

Reducing Barriers

Avoid Question Ping Pong





You

"How do I learn web development?"

*"Uhh, that's a **massive** question - What are you trying to build? For side projects or work? How do you prefer to learn?..."*

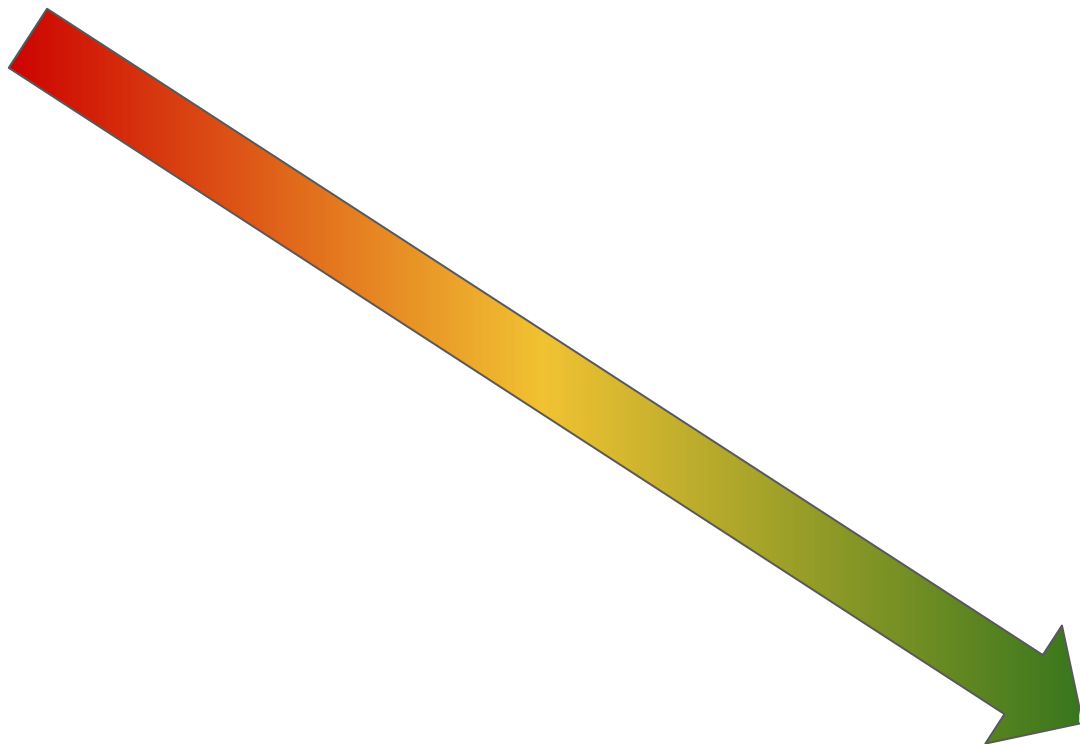
"Oh sorry, here's the context..."



Helper

of
iteration
cycles

Question quality

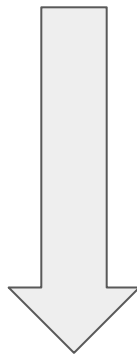


**Predict what helpers will need and
proactively incorporate that.**

Length Balancing

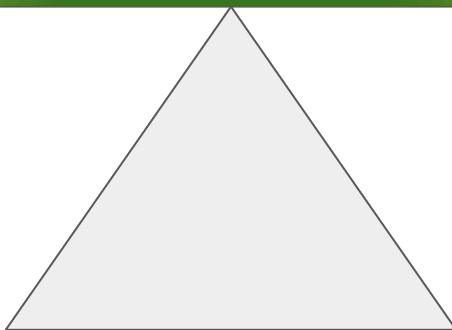


Question is too short, not enough context, follow-up questions needed



We want to be here 

Question is too long, feels like a mental marathon just reading through it 



Make Every Word Packed With Meaning

- Bad length is the classic way questions rob time
 - It's clear and objective
- Sweet spot is 1-2 paragraphs
- **Too short?** Play “question ping pong” in your head
- **Too long?** Rip out things and verify question integrity



^
80

What's the most effective way to switch domains in tech?



Senior Software Engineer [E5] at Meta

2 years ago

Editor's Choice

Learning Quickly

Meta

Senior Engineer

Before I was at Meta, I worked at a networking company for 10+ years. It was very different from Meta as a massive consumer web company. On top of the domain differences, the culture is also very different between these 2 companies.

I'm working on ML at Meta, which is something I'm pretty new to. As a senior engineer who wants to make an impact quickly and make a push for staff, how can I effectively make this domain switch to hit the ground running fast?

 17.6K Views  5 Comments

 Save



"Here's this extremely complicated issue I'm running into: <250+ LINES OF HARD TO READ CODE HERE>

Please help."



"I've been stuck on this nasty launch-blocking crash for 2 days now. After stepping through the end-to-end flow I have isolated the root cause to this specific block of code: [GitHub Link](#)

Thanks for taking the time to read this and happy to hop on a call."

Use links to share deep context



Please Don't Do This: Leading Hello

**WHAT
DO YOU
WANT?**





You



"Hi"

"Hi?"

*"Can you help me with
this?"*

*"Argh this is taking
forever..."*



Helper



You

"Hi, here's my problem."

*"Hello! Here's the
answer."*

*"Awesome, thanks - I
appreciate you 🙏"*



Helper



If you want to be polite, *be efficient.*

Respect their time ⌚ 🙌

no ~~hello~~ |

please don't say just hello in chat

Imagine calling someone on the phone, going *hello!* then putting them on hold... 🙄

❌ Don't do this



Keith 2:15 PM
hi



Tim 2:19 PM
...?



Keith 2:20 PM

Note that Keith could have got his answer minutes sooner, and needn't have kept Tim waiting. In fact, Tim could have started thinking about the question right away!

People who do this are generally trying to be polite by not jumping right into the request, like one would in person or on the phone - and that's great! But it's 2022 and chat is neither of those things. For most people, typing is much slower than

What Have You Tried?

**TRY
HARDER!**

What The Request From A Bad Question Feels Like



If there's 1 thing the tech industry
hates, it's **freeloaders** 🙄



**"PLEASE
DO MY
JOB FOR ME"**



**"I FOUGHT
HARD TO FIGURE
THIS OUT, BUT I'M
STUCK. HELP ME
FINISH THE FIGHT?"**

People help those who are able to help themselves 🤝

Impact is unlocked faster helping capable individuals as they need less hand-holding 🔓

What did Google tell you?



React web-developer roadmap

Google Search

I'm Feeling Lucky

Step 2: Click the Search button

What is this? | [Let Me ChatGPT!](#) | [Skip YouTube Ads!](#) | [Personalized AI Fashion](#)

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What did ChatGPT tell you?

ChatGPT ▾

How to learn React web development?



Learning React for web development can be a rewarding experience as it's a popular library for building user interfaces. Here's a structured approach to help you get started:

1. Understand the Basics of JavaScript and ES6+

Before diving into React, ensure you have a solid understanding of JavaScript, especially ES6+ features like arrow functions, destructuring, classes, and modules.

Resources:

- [MDN JavaScript Guide](#)
- [Eloquent JavaScript](#)
- [JavaScript.info](#)

2. Learn the Fundamentals of HTML and CSS

React is used to build user interfaces, so understanding HTML and CSS is crucial.

Resources:

- [MDN HTML Guide](#)
- [MDN CSS Guide](#)
- [CSS-Tricks](#)



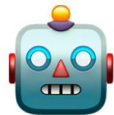
3. Get Started with React



Message ChatGPT



What did company wikis tell you?



Engineering wiki

All things engineering: processes, best practices, setup guides, and more!

Guides & Processes



[Getting started](#)



[Engineering guidelines](#)



[Development lifecycle](#)



[Getting started](#)

Codebase



[Code reviews](#)



[React](#)



[Backend](#)



[Engineering directory](#)

*"I am trying to load an image
in my Android app, and it
shows up blank. Why?"*



"Where is the image coming from?"

"Are you using a library?"

*"Did you declare the **INTERNET**
permission?"*

"I am trying to load an image from this URL using Picasso: tinyurl.com/9xu67

My app has the INTERNET permission, and my emulator can connect to wifi. My ImageView has the proper bounds.

I already tried X, Y, and Z from StackOverflow.

<Links to additional resources with more context>

"I've seen this before! Your image URL is a redirect, so just add this Picasso extension module."



Reduce the problem space.

Technical Questions

RFC Pull Requests

Request for
Comments

**WHAT IS REQUEST
FOR COMMENTS
(RFC)?**

NETWORK ENCYCLOPEDIA

[RFC] Upgrade course completion protocol to modern data model, syncing across surfaces #34

 Open Gear61 wants to merge 16 commits into `master` from `CoursesSupport` 

 Conversation 0

 Commits 16

 Checks 0

 Files changed 21



Gear61 commented 1 minute ago

This PR uses the new `course_completion` collection so lesson completion progress is the same across web, iOS, and Android.

The PR is sort of messy as I wasn't sure of the best place to put the data mutation code. I put it in `PreferencesManager` as that's where the code to maintain user-specific data like history was, but that class feels bloated as it's almost 500 LOC.

Another question I had was around how to handle syncing failure. We update the UI optimistically and make the database update call after. If it fails should we:

- Ignore it and have the UI not match the server state?
- Rollback the UI update and show an error message?
- Retry the database update?

For the rollback, there's a race condition as the user might be on the next lesson when the call fails.

3 In the right sidebar, under "Reviewers," click **Convert to draft**.

Reviewers



octocat



Still in progress? **Convert to draft**

Make sure to add lots of context, particularly by pointing to *problem areas* 🙌

PRs Are Perfect For Adding Context

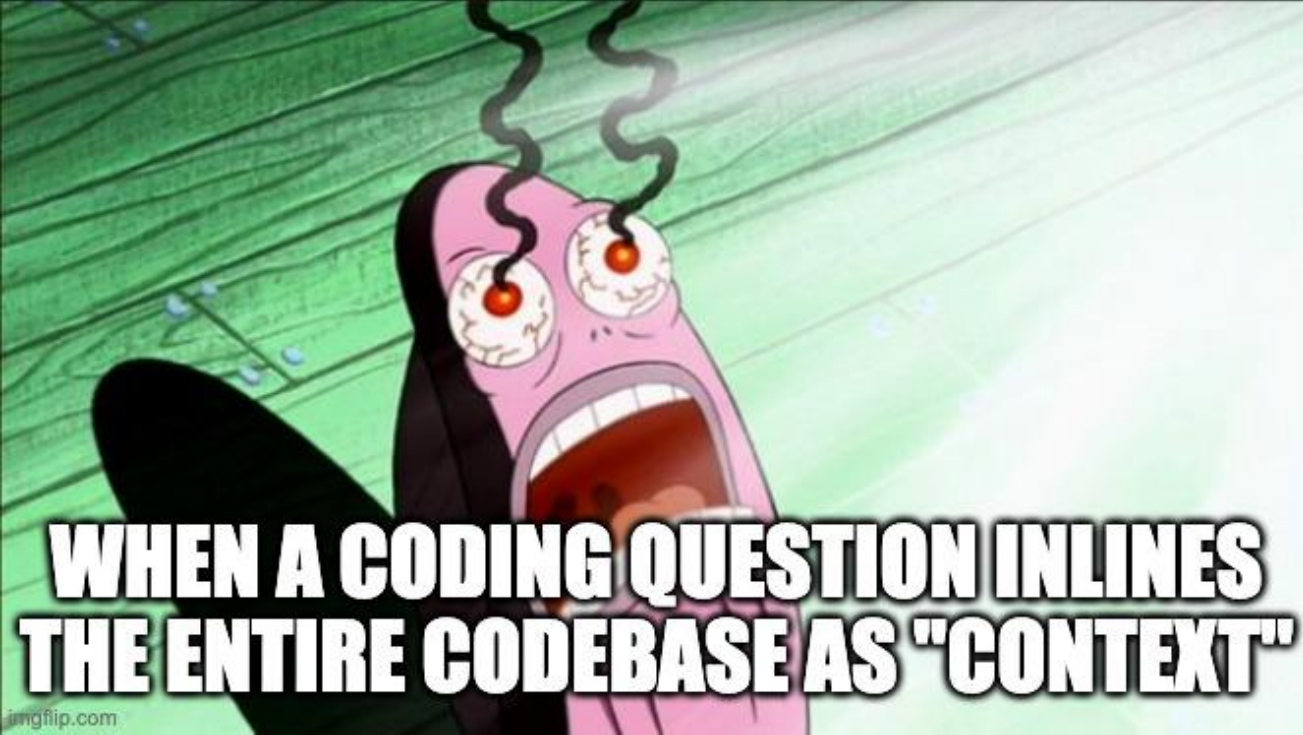
- 1 Title
- 2 Commit names
- 3 Context
- 4 Test plan
- 5 Self-comments



Attaching The Code

Talk is cheap.
Show me the code.

Linus Torvalds



Remember to strike
the balance of not
too much, not too
little ⚖️



"Here's the entire stack trace - Please save me"

```
E Transaction too large, intent: Intent { cmp=com.randomappsinc.techcareergrowth/.learning.LessonActivity (has extras) }, extras size: 216136, icicle size: 0
E Failed to find sync for id=0
E eglQueryContext 32c0 EGL_BAD_ATTRIBUTE
E tid 10443: eglQueryContext(2122): error 0x3004 (EGL_BAD_ATTRIBUTE)
E FATAL EXCEPTION: main
Process: com.randomappsinc.techcareergrowth, PID: 11061
java.lang.RuntimeException: Unable to start activity ComponentInfo{com.randomappsinc.techcareergrowth/com.randomappsinc.techcareergrowth.home.MainActivity}: java.lang.IllegalArgument
    at android.app.ActivityThread.performLaunchActivity(ActivityThread.java:3645)
    at android.app.ActivityThread.handleLaunchActivity(ActivityThread.java:3782)
    at android.app.servertransaction.LaunchActivityItem.execute(LaunchActivityItem.java:101)
    at android.app.servertransaction.TransactionExecutor.executeCallbacks(TransactionExecutor.java:135)
    at android.app.servertransaction.TransactionExecutor.execute(TransactionExecutor.java:95)
    at android.app.ActivityThread$H.handleMessage(ActivityThread.java:2307)
    at android.os.Handler.dispatchMessage(Handler.java:106)
    at android.os.Looper.loopOnce(Looper.java:201)
    at android.os.Looper.loop(Looper.java:288)
    at android.app.ActivityThread.main(ActivityThread.java:7872) <1 internal line>
    at com.android.internal.os.RuntimeInit$MethodAndArgsCaller.run(RuntimeInit.java:548)
    at com.android.internal.os.ZygoteInit.main(ZygoteInit.java:936)
Caused by: java.lang.IllegalArgumentException: Invalid document reference. Document references must have an even number of segments, but users has 1
    at com.google.firebase.firestore.DocumentReference.forPath(DocumentReference.java:81)
    at com.google.firebase.firestore.CollectionReference.document(CollectionReference.java:104)
    at com.randomappsinc.techcareergrowth.home.UserSyncManager.startSyncing(UserSyncManager.kt:53)
    at com.randomappsinc.techcareergrowth.home.MainActivity.onCreate(MainActivity.kt:94)
    at android.app.Activity.performCreate(Activity.java:8305)
    at android.app.Activity.performCreate(Activity.java:8284)
    at android.app.Instrumentation.callActivityOnCreate(Instrumentation.java:1417)
    at android.app.ActivityThread.performLaunchActivity(ActivityThread.java:3626) <12 more...>
```



"I stepped through the code and isolated the part of the stack trace pointing to the directly relevant code - Here it is."

```
Caused by: java.lang.IllegalArgumentException: Invalid document reference. Document references must have an even number of segments, but users has 1
    at com.google.firebase.firestore.DocumentReference.forPath(DocumentReference.java:81)
    at com.google.firebase.firestore.CollectionReference.document(CollectionReference.java:104)
    at com.randomappsinc.techcareergrowth.home.UserSyncManager.startSyncing(UserSyncManager.kt:53)
```

"I wrote the API call, but the items aren't rendering in the list. Can someone help?"

<INLINED 250+ LINES OF MY SPAGHETTI CODE HERE>"



"I'm making the API call with Retrofit (verified the response) and passing the items to the adapter, but the list still isn't rendering anything. Any ideas?"

I polished up my code and put it in an RFC PR here (test plan and relevant logs included): [RFC PR Link](#)"

Links are great for code 



Target Domain Experts





Helper

***“Am I the right
person to
answer this
question?”***

"Hi team, I am trying to get the rewards microservice to return timestamps, but it always returns 0. Why?"



"Hi team, I am trying to get the rewards microservice to return timestamps, but it always returns 0. Why?"

cc [@Alex](#), [@Charlie](#) as it looks like they most recently worked on rewards"



How To Find Domain Experts?

- 1 Blame code
- 2 Ask around (especially your manager)
- 3 Blame wikis
- 4 Look at ownership charts

Favors

Put Them On A Pedestal



Make People Feel Special

- The people who get asked for favors are usually special
 - So make them feel that way!
- Everyone loves praise, and it's **free** to generate
- *Sharing positive energy goes a long way*



"Can you mentor me?"



"I just wanted to tell you that your AI research work is amazing! I went through your talk last week and I love your idea of using multi-token prediction instead of next-token prediction to train LLMs..."

I'm looking to transition into MLE and would love to chat sometime to learn more from you. 😊"



Give Them An Out



Make It Easy For Them To Say “No”

- The people asked for favors tend to be popular
 - It's a slog reading through demands
- Altruistic people tend to feel bad if they can't help
- Give them “permission” to not feel bad at all if they can't help - *Ooze kindness*



"Can you mentor me?"



*"I'm a huge fan of your ML work...
<Pedestal placing>*

*I'm looking to transition into MLE and
would love to chat sometime to learn
more from you.*

***No worries at all if you can't - I'm sure
your schedule is constantly packed! The
advice you post is already so helpful 😊***



Make Fulfillment Easy

[illegible]

"Can you mentor me?"

What *exactly* do you want and how can you reduce their burden to deliver it?



"I just wanted to say that I'm a huge fan... <Pedestal placing>

*I'm looking to transition into MLE and would love to chat. **Does 1PM Monday or Thursday work for you?***

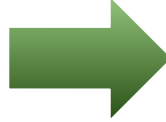
<Give them an out>"



"I just wanted to say that I'm a huge fan... <Pedestal placing>

I'm looking to transition into MLE and would love to chat. Does 1PM Monday or Thursday work for you?

<Give them an out>"



"<Pedestal placing>

*<Context>. Does 1PM Monday or Thursday work for you? **If they don't, I can make 1PM - 5PM work on any day by rearranging my calendar.***

<Give them an out>"



"Can you review my code?"



Call out problem areas in self-comments

"Can you review my resume?"



Extract specific sections and ask for feedback on that

"Can you tell me what the interview is like?"



Include a list of LeetCode problems and ask to verify

Follow The Golden Rule

Treat others THE way
You want TO BE Treated



**You get value from someone by
adding similar value to them.**

"Can anyone here review my resume?"



"I've been having trouble getting interviews - Do any of you see red flags with my resume? <PDF LINK>

Happy to review anyone else's resume as well - Just drop a PDF and I'll add comments ASAP (I can even hop on a call to explain)!"



"Can you review my code?"



Review some of their PRs

"Can you tell me what the interview is like?"



Offer to share your experiences with other companies

"Can you refer me?"



Offer to refer them to companies in your network

But what if we can't add an equivalent amount of value to them? 🤔

Step 1: Do research on the person

Step 2: Find a thing they care about

Step 3: Help that thing (be creative)

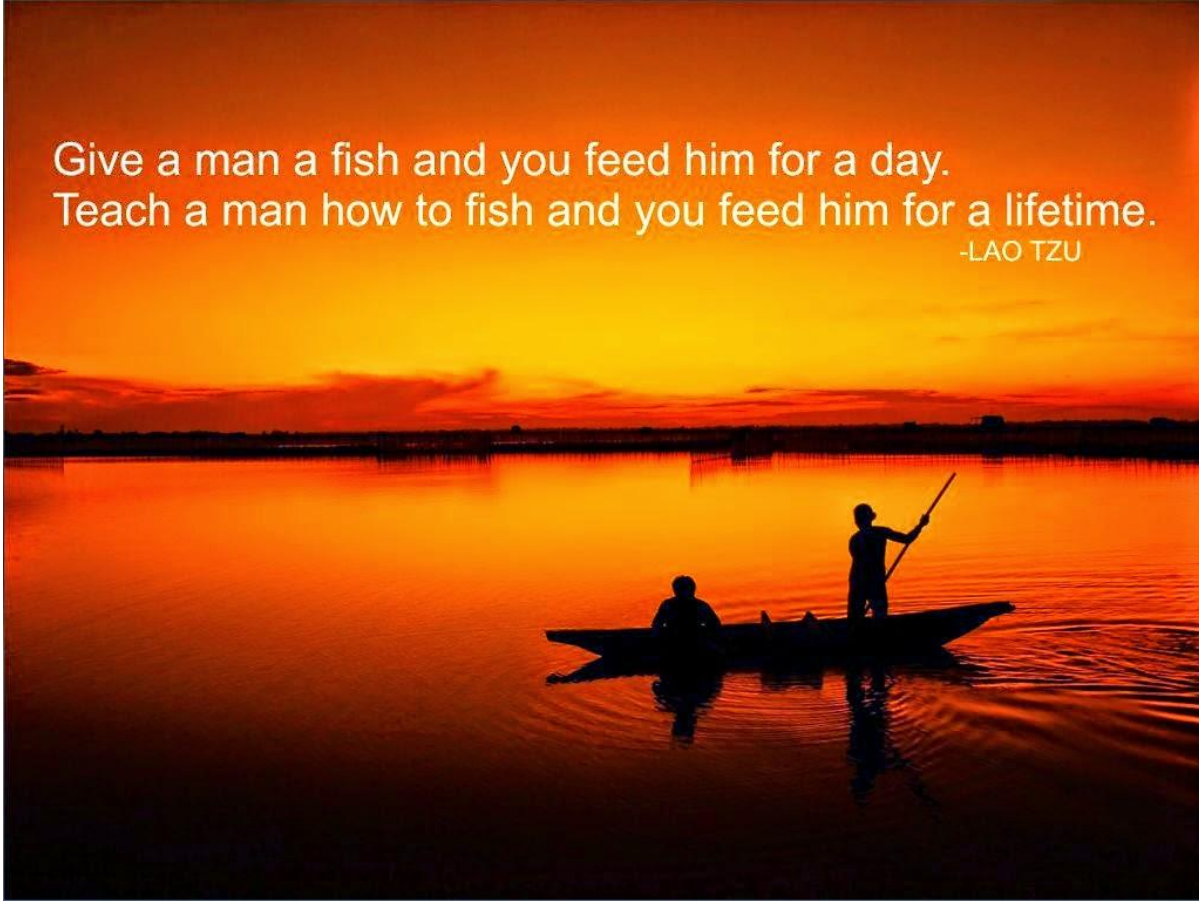


Stick The Landing

Learn To Fish

Give a man a fish and you feed him for a day.
Teach a man how to fish and you feed him for a lifetime.

-LAO TZU





You

"Hi, I have this issue - Can you help?"

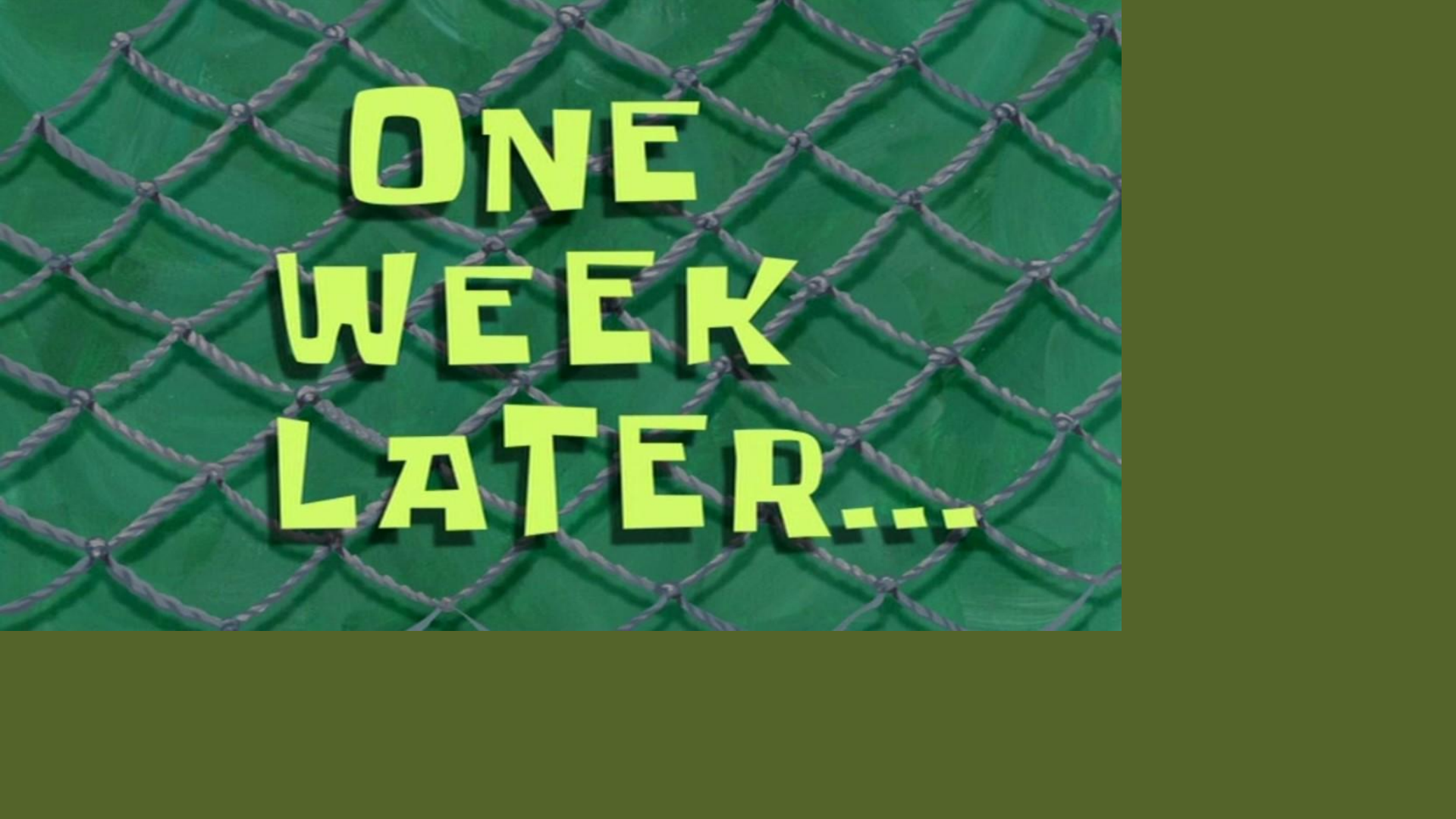
"If you apply fix ABC, the issue should go away."

"It worked - Thanks!"



Helper





**ONE
WEEK
LATER...**



You

*"Hi, I have this other issue -
Can you help?"*



Helper

*"This issue is 90%
similar to their issue
last week, can they
not do anything...?"*



You



"Hi, I have this issue - Can you help?"

"If you apply fix ABC, the issue should go away."

"Thanks for the help! Sorry to bother you more, but can you explain how you got ABC?"

I want to learn your process, so I can unblock myself in the future and not take more of your time 🙏"



Helper

**The point of asking questions is to
build *mastery*, not knowledge.**

Deep Thanks





You

"Hi, I have this issue - Can you help?"

"If you apply fix ABC, the issue should go away (let me know if it doesn't).

For the future, you can use [this stack trace analysis tool](#) to quickly root cause the breaking line of code - That's how I got the fix."

"Thanks"



Helper





You



"Hi, I have this issue - Can you help?"

"<Beautiful answer with fix, a follow-up if necessary, and the explanation>"

"Thank you for the help! I'm so grateful for everything you teach me, and I always write it all down so I never forget. You're too good to me 😭"



Helper

Ways To Give Deep Thanks 🙌

- 1 Thank them in-person
- 2 Thank them in public
- 3 Thank them on a recurring basis
- 4 Thank their manager
- 5 Buy them something 🧋 ☕ 🍫

Showing gratitude is a skill, not a formality. Become amazing at it.

Conclusion

Give More Than You Take

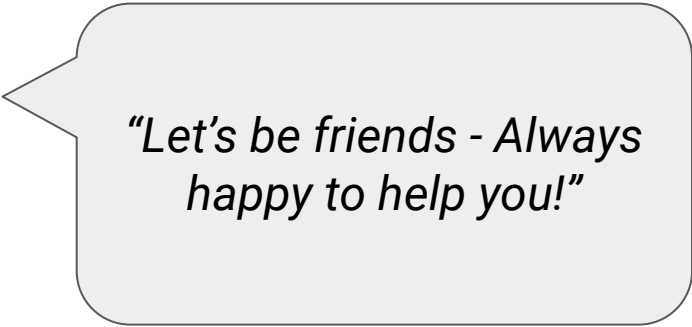




Altruistic
Helper

Selfish
Leech

`!opposites.attract()`



*"Let's be friends - Always
happy to help you!"*

Altruistic
Helper

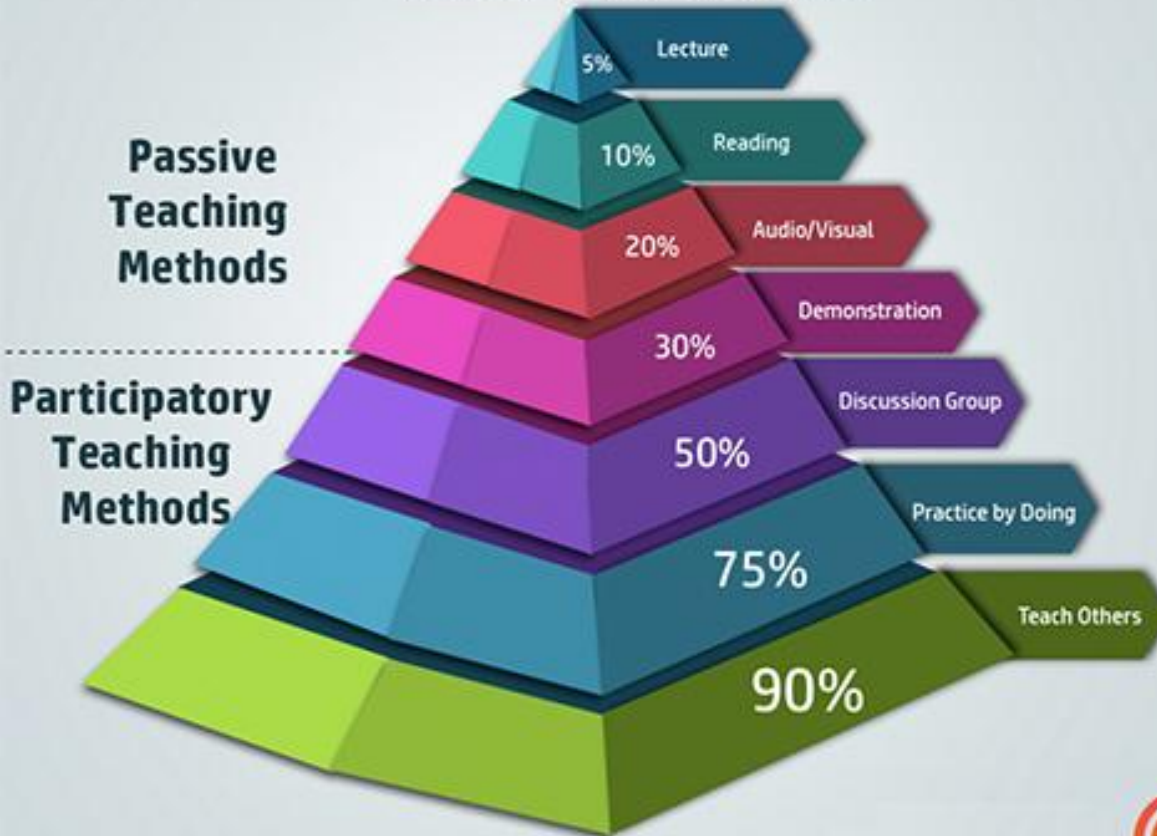
Altruistic
Helper

like.attract(like)

**Answer the questions of others to get
answers on your own questions.**

THE LEARNING PYRAMID

KNOWLEDGE RETENTION RATES



Adapted from National Training Laboratories, Maine



Teaching is also one of the best ways to learn! 🧑🏫



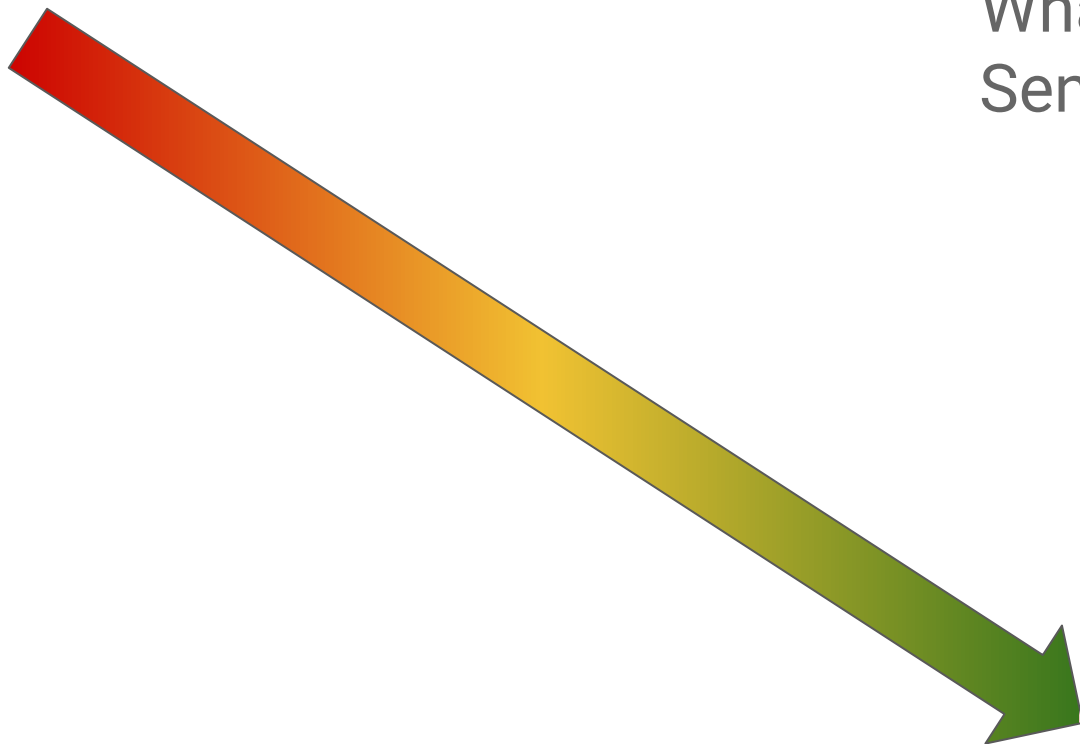
How You Should Evolve



What People *Think*
Seniority Means ❌

of
questions
asked

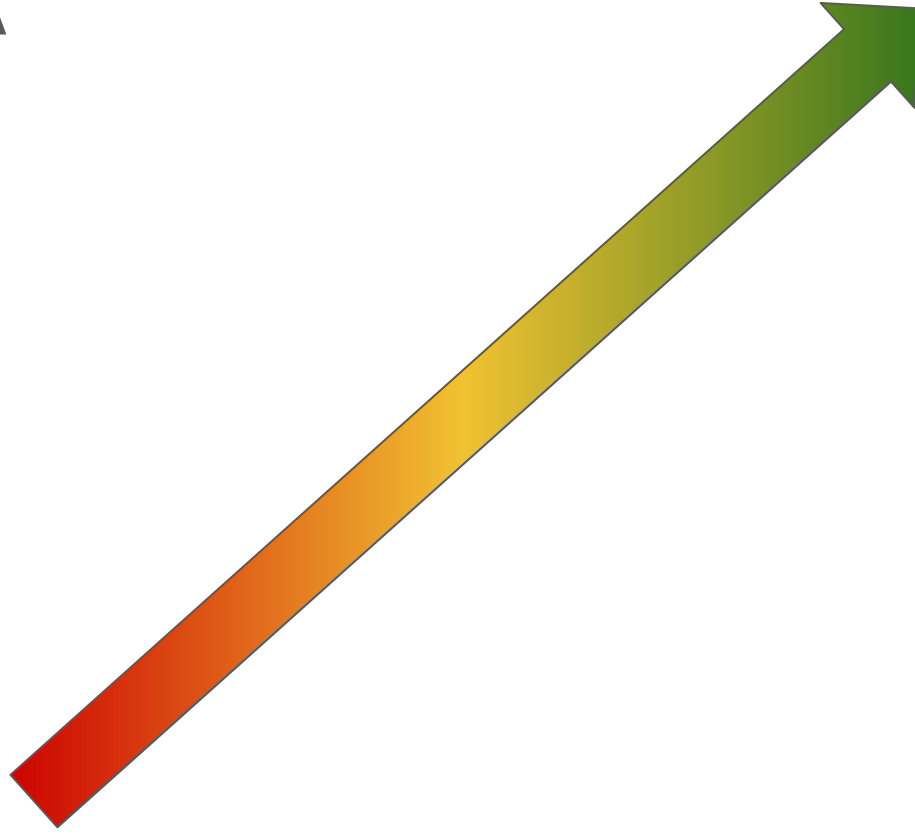
Seniority



Question
quality

Seniority

What Seniority
Actually Means ✓



Level 3: *"I know this space inside and out: Here's a bunch of good solutions - Let's find the absolute best"*

Senior+



Level 2: *"I know how this space works - I came up with solutions but need help finding a good one"*

Mid-Level



Level 1: *"I don't know anything"*

Junior

Go Deeper: Follow Through

**TRANSFORM
KNOWLEDGE
INTO VALUE**



Put yourself in the helper's shoes.

There are no stupid questions.

Reduce the problem space.

Add value to get value.

Give deep thanks.



Professor Alex's Homework For You

- ❑ Give deeper thanks to someone who's helped you recently
- ❑ Long-term: Figure out a domain to be a “question sponge”
- ❑ Find 1 area that you can ask a question about
- ❑ Optimize the last 5 questions you have asked
- ❑ Answer 5 questions and reflect

Title

Is asking 5 questions per day too much?

Body

Add more details - the more information you share, the better the community can help you!

H **B** *I* S ☰ ☷ “ </> 🔗



This is really important
if you're asking
questions in-person
(virtual or in-office) 

Effective Communication For Engineers



LEVEL UP YOUR CODE QUALITY



A course by:

ALEX CHIOU



Writing better code
(especially with PR
“dressing”) greatly
levels up your technical
questions 

**Level Up Your Code Quality As A
Software Engineer**

thank you 